



# Research on Autonomous Robotic Recognition and Disassembly of Bolts

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**Abstract.** In today's global manufacturing industry transitioning toward low-carbon and high-efficiency development, the recycling and remanufacturing of high-value equipment has become increasingly critical. As bolts are the most commonly used fasteners in the majority of equipment, research into autonomous bolt recognition and disassembly technologies plays a key role in reducing labor costs during the remanufacturing process. Initially, a robotic disassembly system and a circular fastener equipped with an upper guard plate were designed. Subsequently, given the difficulty of acquiring large-scale bolt datasets, a comparative analysis was conducted by training YOLO-based recognition models using both real-world datasets derived from the circular fastener and synthetic datasets. The training results demonstrate that properly processed synthetic datasets can be utilized to train models with satisfactory recognition capabilities, effectively reducing the reliance on collecting extensive real-world data. Furthermore, based on the bolt recognition algorithm, disassembly target localization and autonomous disassembly planning were implemented, followed by physical experiments in real-world environments. The experimental results validate the high success rate and accuracy of the proposed recognition and planning framework.

**Keyword:** synthetic datasets · visual recognition · bolt disassembly

## 1 Introduction

In the context of global decarbonization efforts, remanufacturing has been acknowledged as an effective technology that enhances energy and material utilization while reducing environmental emissions at low cost [1]. The disassembly of products is a critical step in the remanufacturing process and also a labor-intensive stage. In recent years, the rapid technological advancement in machine vision and robotics has facilitated the automation of this process. A critical challenge in bolt disassembly lies in the identification and localization of bolts. Existing publicly available datasets suffer from issues such as limited quantity and inadequate annotations, while manual labeling of relevant datasets would require significant human labor. Recent advancements

in computer technology have made the application of synthetic datasets a viable solution. A. I. Károly and P. Galambos [2] reviews tools for synthetic dataset generation in object segmentation and demonstrates a Blender-based workflow, highlighting key technical considerations. Denninger M et al. [3] have developed a modular procedural pipeline named BlenderProc based on Blender, designed to generate realistic-looking images for training convolutional neural networks. Zhu et al. [4] introduce SIP-17, a synthetic-only dataset of 66 000 industrial part images, and demonstrate that domain randomization significantly narrows the sim-to-real performance gap in classification tasks. Rawal et al. [5] addresses the challenge of visual recognition for complex assemblies in industrial production by introducing a scalable, pipeline-driven approach for generating synthetic training data. Beyond the use of synthetic datasets, numerous studies have directly focused on the identification and localization of bolts. Zhao Y et al. [6] developed a dataset called NPU-BOLT and made it publicly available for researchers to use and further develop. Liu Q et al. [7] proposed a two-stage detection framework for structurally damaged screws and used a linear regression model with reflection features, allowing the detection framework to operate under uneven lighting conditions. Deng W et al. [8] proposed a predictive exposure control method that enables real-time acquisition of high-quality images in low-light conditions.

1.1 Hardware Design

Accurate bolt recognition and localization capabilities are essential for robotic systems in bolt disassembly tasks. Due to mutual occlusion among internal components of the circular fastener, the RGB-D camera should be mounted on the robotic arm’s end-effector to guide the disassembly tool (Table 1 and Fig. 1).

When applying a robotic arm to disassemble bolts, it is necessary to determine the relative positional relationship between the tool and the bolt. The recognition system can obtain the bolt’s position relative to the camera, while hand-eye calibration provides the relative pose between the camera and the robotic arm’s end-effector. Additionally, the dimensions of the fixture and disassembly tool can be used to derive the transformation relationship between the disassembly tool and the robotic arm’s end-effector. Finally, by expressing these relationships as transformation matrices and multiplying them, the transformation matrix between the bolt and the disassembly tool can be calculated. This matrix guides the robotic arm in performing the bolt disassembly task.

Table 1. Hardware

| Name                      | Model                | Quantity |
|---------------------------|----------------------|----------|
| Collaborative Robotic Arm | Dobot CR5            | 1        |
| RGB-D Camera              | Intel RealSense D415 | 1        |
| Electric Torque Wrench    | BOSCH Go 3           | 1        |
| Circular Fastener         | N/A                  | 1        |

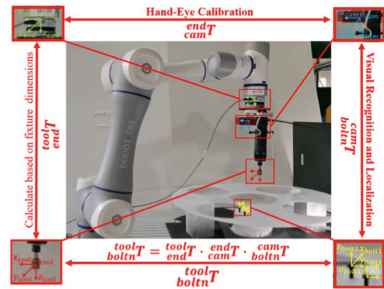


Fig. 1. Disassembly System

1.2 Disassembly Process Planning

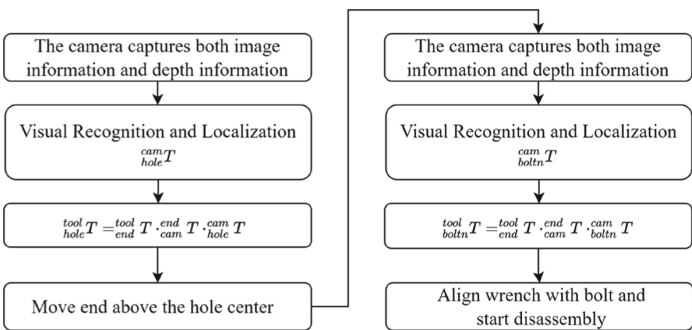


Fig. 2. Disassembly Process Planning

To reduce bolt recognition and positioning errors, thereby improving the success rate, the program will first identify the positions of circular holes on the upper guard plate. The robotic arm will then move the imaging center of the camera above the center of these circular holes before recognizing and positioning the bolts, after which dismantling will commence. The circular hole recognition and positioning algorithm on the upper plate employs techniques such as edge detection and elliptic matching. As this is not the focus of the paper, the implementation details of this algorithm will not be discussed in subsequent sections (Fig. 2).

2 Visual Recognition and Localization

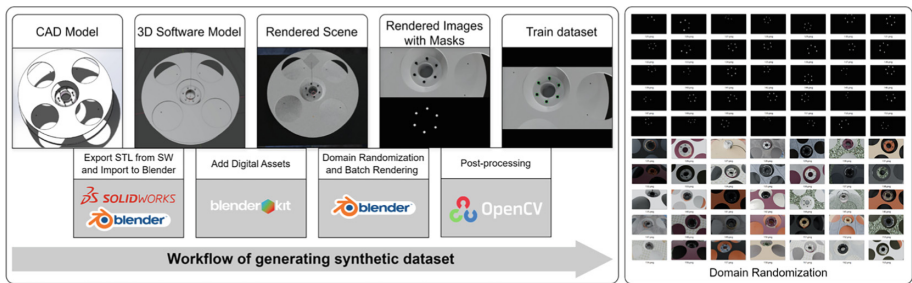
To acquire precise bolt poses for operational guidance, the vision system must reliably identify the central region of bolt heads and compute their 3D positions relative to the camera frame.

A straightforward approach for bolt recognition involves training YOLO-based detection models using dedicated datasets. However, publicly available datasets are limited in quantity, and their bounding boxes typically encompass the entire bolt rather than

focusing on the bolt head's central region, leading to suboptimal localization accuracy. Furthermore, directly collecting and annotating real-world datasets would demand substantial human effort. To address these limitations, this study explores the feasibility of leveraging synthetic datasets. A circular fastener equipped with an upper guard plate was designed and utilized as the basis for experimental validation.

## 2.1 Comparative Analysis of Synthetic and Real-World Datasets

Blender is an open-source 3D animation software with active community support, a rich plugin system, and a Python API that allows for easier batch generation of required images. Therefore, Blender is highly suitable for synthetic dataset generation projects.



**Fig. 3.** Workflow of generating synthetic datasets

Figure 3 illustrates the workflow for generating the synthetic dataset. First, the SolidWorks model files were converted to STL format. These STL files were then imported into Blender, an open-source 3D modeling and rendering software. Next, free assets from the BlenderKit digital asset library were used to apply HDR environment lighting and material textures to the models. The Blender Node Editor and custom Python scripts were employed to batch-render RGB images and generate precise masks corresponding to bolt head regions. Finally, the OpenCV library was used for mask post-processing to produce annotation label files. The rendered RGB images and processed label files together comprised the training dataset. To enhance the diversity of the synthetic dataset during Blender rendering, materials for the bolts and other objects were randomized; bolts were randomly missing, variations in height, rotations around the Z-axis, and size changes; environmental brightness was varied randomly; and camera positions and field of view were also randomized. In total, 502 synthetic images were generated in this experiment.

A real dataset of 360 images was used for comparison, of which 72 images were reserved as the validation set common to both the synthetic and real datasets. The training steps and configurations for both training sets were identical. The validation set served only to evaluate model parameters and did not affect the training process. The synthetic dataset exceeds the real dataset in size because once the rendering pipeline is implemented, generating large volumes of data requires no manual effort—an inherent advantage of synthetic data.

Training used the YOLOv11n base model for a total of 300 epochs. Because there is only a single detection target, Average Precision and mean Average Precision (mAP) are equivalent. In this context, mAP50 refers to the area under the Precision–Recall curve at an Intersection over Union (IoU) threshold of 0.5. From Table 2, comparing models trained on synthetic and real datasets shows that both achieved similar accuracy at an IoU threshold of 0.5. mAP50–95 is the mean of mAP values computed at IoU thresholds ranging from 0.5 to 0.95 in steps of 0.05, which is a very strict metric. Even when the IoU threshold is set very high, the performance of the model trained on the synthetic dataset remains very close to that of the model trained on the real dataset.

In this study, using an RTX 4060 Ti GPU, each image and its corresponding mask can be rendered in approximately 15 s, while the time required for mask post-processing and label generation is negligible. In contrast, collecting a diverse real dataset to avoid over-fitting requires multiple sessions at different times and locations. Although the dataset contains only one class (bolts), each image takes around 30 s to annotate to ensure high quality. Therefore, creating a large real dataset is time-consuming and labor-intensive. If ultra-high accuracy is not required, synthetic datasets offer a practical alternative.

**Table 2.** Comparison of Training Accuracy between Real and Synthetic Datasets (IoU = 0.5)

| Dataset           | mAP50   | mAP50–95 | Precision | Recall  |
|-------------------|---------|----------|-----------|---------|
| Real dataset      | 0.99498 | 0.88356  | 0.99520   | 1.00000 |
| Synthetic dataset | 0.99481 | 0.83891  | 0.99264   | 0.99532 |

**2.2 Bolt Localization Algorithm**

Table 3 presents the pseudocode of the bolt recognition and localization algorithm, whose inputs include the RGB image (Image), depth map (DepthFrame), and camera intrinsic parameters (CameraParameters).

**3 Experiment**

**3.1 Camera Intrinsic Calibration and Hand-Eye Calibration**

To achieve sufficient accuracy, camera intrinsic calibration is essential; otherwise, distortions in the camera parameters may introduce errors during hand-eye calibration, resulting in biased outcomes. After intrinsic calibration, the calculated pose of the calibration board becomes more precise, establishing a reliable foundation for calibrating the hand-eye matrix.

Equation (1) represents the camera intrinsic calibration results, where  $f_x$  and  $f_y$  denote the focal lengths along the X-direction and Y-direction, respectively, measured in sensor pixel units. The tangential distortion coefficients  $p_1$  and  $p_2$  are employed to correct distortions caused by the misalignment between the lens and sensor during

**Table 3.** Bolt Detection Pseudocode

|                                                                                 |
|---------------------------------------------------------------------------------|
| Algorithm: DetectBolt(Image, DepthFrame, CameraParameters)                      |
| # Use the model to detect bolts and obtain bounding boxes and confidence scores |
| 1.BoltBox, Confidence = Best_model(Image)                                       |
| # Calculate the center pixel coordinates from the bounding boxes                |
| 2.CenterPixels = Center(BoltBox)                                                |
| # Align the depth frame to the color image using camera parameters              |
| 3.AlignDepthFrame = AlignDepth2Color(DepthFrame, CameraParameters)              |
| # Retrieve depth values at bolt center pixels from the aligned depth frame      |
| 4.DepthOfBolts = AlignDepthFrame(CenterPixels)                                  |
| # Compute 3D positions in camera coordinates and return results                 |
| 5.BoltPositions = ComputePositons(DepthOfBolts, CenterPixels, CameraParameters) |

imaging. Notably, radial distortion was found negligible and thus remained uncalibrated. The calibrated hand-eye matrix is presented in Eq. (2). There are many factors that affect the accuracy of hand-eye calibration, such as errors in the camera intrinsic calibration, positioning errors of the robot arm, and the choice of robot poses during calibration [9]. In subsequent experiments, it was found that when using an eye-in-hand setup, the rotation matrix had a significant impact on the camera's recognition and localization accuracy: even a small error in this matrix could lead to substantial localization errors when detecting objects. To reduce this error, the camera fixture should have sufficient stiffness, high positioning accuracy, and as short a cantilever structure as possible to minimize fixture deformation.

$$\begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1370.18 & 0 & 971.13 \\ 0 & 1370.10 & 537.82 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

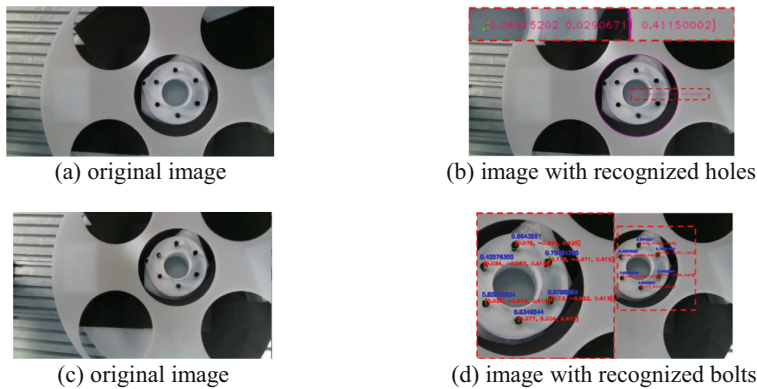
$$p_1 = 0.0635 \quad p_2 = 0.0838$$

$${}_{cam}^{end}T = \begin{bmatrix} 0.7111 & 0.7030 & -0.0058 & -38.8912 \\ -0.7028 & 0.7111 & 0.0179 & 10.6067 \\ 0.0167 & -0.0086 & 0.9998 & 89.9000 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

### 3.2 Recognition Experiment and Disassembly Accuracy Verification

Figure 4(a) and Fig. 4(b) illustrate the experimental results of circular hole recognition on the top plate. The purple circles represent the hole contours matched by the algorithm, while the center coordinates are determined based on the matched contours and depth map.

The practical results of the bolt recognition model are illustrated in Fig. 4(d). Green bounding boxes represent the detected bolts, blue numbers indicate confidence scores, and red numbers denote the coordinates of the bolts in the camera coordinate system, calculated based on the detection results.



**Fig. 4.** Circular hole and bolts recognition experiment images

In order to verify the accuracy of the bolt positions obtained from the recognition process and to assess the feasibility of the disassembly experiment, this paper measures the end-effector position error of the robotic arm when the wrench and bolt head are aligned using the transformation matrix and camera information, and compares it with the manual teaching alignment. The results are shown in Table 4. The XYZ axes represent the three coordinate directions of the robot’s base coordinate system, and the robot’s three rotational angles remain constant, ensuring that the Z-axis of the wrench is always pointing straight down during the movement.

The average errors in the X and Y directions are both less than 1mm, indicating high accuracy. Although the error in the Z-axis is larger, the bolt sleeve has a certain length, which allows for some tolerance. As long as the error is not too large, it does not affect the disassembly of the bolt.

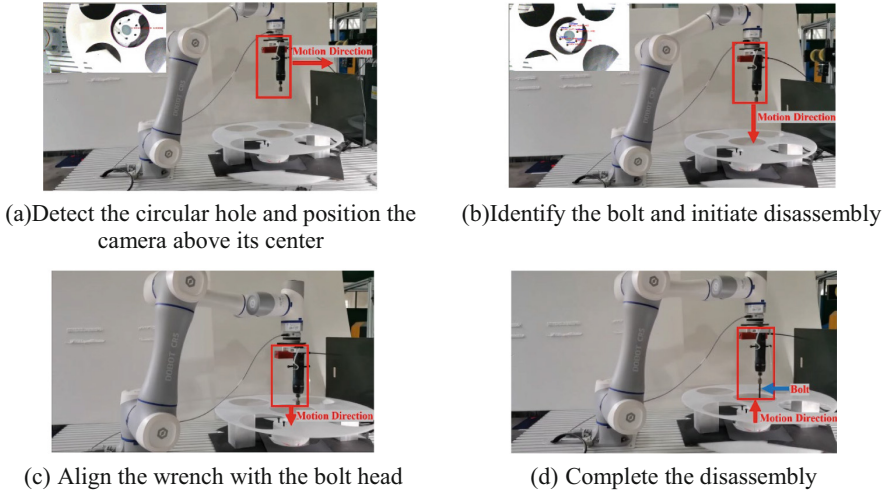
**Table 4.** Disassembly Accuracy Verification

| Error direction   | X     | Y     | Z     |
|-------------------|-------|-------|-------|
| Average error /mm | 0.823 | 0.798 | 1.552 |

**3.3 Disassembly Experiment**

Figure 5 shows the bolt disassembly experiment diagram.

In Fig. 5(a), the upper left corner displays the recognition and positioning results of the circular hole on the upper plate. After recognition, the camera is aligned to the center of the circular hole to reduce bolt recognition errors. In Fig. 5(b), the upper left corner shows the recognition and positioning results of the bolt. Once identified, the robotic arm is controlled based on the bolt’s position to initiate disassembly.



**Fig. 5.** Disassembly experiment images

## 4 Conclusion

This study designs corresponding disassembly devices and a circular fastener. Subsequently, a synthetic dataset was generated based on this fastener, and a real dataset was collected. Models trained on synthetic and real datasets demonstrated similar accuracy when evaluated at an IoU threshold of 0.5. This demonstrates that synthetic datasets, when properly processed, can approximate the performance of real datasets in scenarios where acquiring large-scale real data is challenging, thereby significantly reducing reliance on real-world data. A bolt recognition model was then developed to establish a bolt positioning procedure, followed by hand-eye calibration experiments and accuracy verification. The bolt recognition and positioning system achieved an average XY-direction error of less than 1 mm and a Z-direction error below 2 mm. Finally, comprehensive disassembly experiments demonstrated successful bolt identification, positioning, and removal, validating the robustness and accuracy of the proposed framework.

**Acknowledgement.** This paper is supported by the Key Research and Development project of Hubei Science and Technology Plan [Grant No. 2024BAB055].

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